SOVEREIGN INVESTOR DASHBOARD · SLOVAKIA & CZECHIA

Audit, Test Results & Capability Map

Bugs and problems ranked by severity · every feature mapped · open suggestions · your manual to-do list with step-by-step guides

3,734	0	29	4
TESTS PASSED	TESTS FAILED	FINDINGS	HIGH SEVERITY

Review date: 6 November 2026 · Code reviewed: `main` at commit `79e8edd` (after PR #72) · Read-only review: nothing in the app's code was changed. The real database and real listing data were never touched.

Contents

1. Every finding at a glance (ranked)
2. Everything the dashboard can do — the full capability map
3. Test and check results
4. Findings in detail, ranked by importance and severity
5. What happened to the previous review (5 Nov 2026)
6. Suggestions: open ones from the previous review, and new ones
7. What you need to do manually — step by step, with resources
8. How this review was done, and sources

1. Every finding at a glance

Overall: the code is healthy. All 3,734 tests pass, the linter is clean, and every one of the 11 tabs opens in a real browser without an error. No crash in normal use turned up. The problems are in **what the model assumes**: tax rules, the title-deed (LV) screen and how the two ownership structures are compared. These are the places where a wrong number can lead to a wrong buying decision, so they rank highest.

Severity. HIGH can give a wrong buy/skip decision or a wrong number in normal use. MEDIUM gives weaker or misleading results in some cases, or is a risk that grows over time. LOW is cosmetic, a rare edge case or housekeeping. INFO is a note, nothing to fix. **Effort:** S = under half a day · M = a few days · L = a week or more. ✔ = reproduced with a test script during this review.

#	Severity	Finding	Where	Effort
1	HIGH	✔ LV screen misses a **transfer ban** (“zákaz scudzenia / zákaz prevodu / obmedzenie prevodu”) on Slovak title deeds, so such a flat can come out as clean	`modules/lv_screen.py` rules	S
2	HIGH	2026 Slovak personal income tax table is out of date: it models 2 bands (19/25 %, €41,445) where 2026 has 4 (19/25/30/35 %, from about €43,983). The model also ignores your other income.	`config.py` `TAX_YEAR_RULES`	S–M
3	HIGH	Tax tables expire on **1 January 2027** (8 weeks away). There are no 2027 tables for either country. Still open from the previous review (its H4).	`config.py`, `markets.py`, property-tax engines	M (yearly)
4	HIGH	✔ The common **statutory lien under § 15 of Act 182/1993**, in favour of the other flat owners in the building, is treated as a private lien. Without a Claude key, every flat that has it is REJECTED.	`modules/lv_screen.py`	S
5	MEDIUM	✔ The s.r.o. vs personal comparison is not like for like. The s.r.o. borrows 70 % against personal 80 %, so its “saving” is mostly the extra €11–15k of your own cash, counted as if it were free.	`engine/financial.py` `analyse()`	S
6	MEDIUM	Personal tax doesn't deduct mortgage interest. Sources say interest is deductible for § 6(3) rental, so personal tax is overstated.	`personal_rental_base()`	S + adviser
7	MEDIUM	Cash-on-cash, net yield and total ROI are always the s.r.o.'s figures, even when Personal is the recommended structure (most listings).	`analyse()` key metrics	S
8	MEDIUM	The What-If “Hold (years)” slider changes the IRR but not the “Best structure” verdict, which always assumes a 10-year hold.	`what_if.py`, `analyse()`	S
9	MEDIUM	✔ Czech towns outside the rent index (Kladno, Mladá Boleslav, Opava, Frýdek-Místek, Karviná, Teplice, Most …) stay unscored until 5 of their own rentals are scraped	`engine/rent_cz.py`	S–M
10	MEDIUM	Without an Anthropic key, the LV decision rests on keyword rules alone. Findings 1 and 4 show how far those can miss.	`modules/debt_bot.py`	config
11	MEDIUM	Backups sit on the same disk as the database, with no off-site copy	`modules/backup.py`	S (manual)

12	MEDIUM	Owner names and full LV texts in `cadastre_cache` are kept forever, with no retention limit (GDPR)	database	S
13	MEDIUM	Seven Slovak towns' property-tax rates still use the €1.35/m² fallback because their websites block automated reads	TAX RATES tab	S (manual)
14	LOW	When a scraper hits the 30-minute timeout, the Python child is killed but Playwright's browser processes may be left running	`dashboard/actions.py`	S
15	LOW	✔ `calc_mortgage` divides by zero at a 0 % rate or a 0-year term, so a 0 % family loan can't be modelled	`engine/financial.py:186`	S
16	LOW	What-If shows an IRR of “0.0 %” when the IRR can't be worked out, instead of “—”	`what_if.py`	S
17	LOW	✔ The triage table shows a grey “None” in the Max 🟡 / Max 🟢 / To 🟡 columns when a listing has no benchmark	`dashboard/tabs/triage.py`	S
18	LOW	✔ Streamlit's “Deploy” button is still visible (previous idea 13)	`.streamlit/config.toml`	S
19	LOW	`logs/scheduler.log` is never rotated and grows forever	`scheduler.py`	S
20	LOW	The Docker image runs as root and ships a compiler it doesn't need	`docker/Dockerfile`	S
21	LOW	Slovak contract draft wording: “dle” is Czech, the § 56a reference needs checking, it implies a notarial deed is required, and it uses the portal's ad title as the property description	`modules/contract_draft.py`	S + notary
22	LOW	✔ Any creditor whose name contains “banka” is cleared as a bank, even an s.r.o. (Slovak banks must be a.s.)	`modules/lv_screen.py` `_BANK_RE`	S
23	LOW	Each saved listing opens a new database connection and re-runs 3 schema checks, which slows big scrapes	`database/listings.py` `upsert_listing`	S
24	LOW	The price-history query sends every listing ID in one `IN (…)` list, with no batching (the title-deed queries batch at 500)	`database/prices.py:90`	S
25	LOW	A rent override of 0 means “no override” (`rent_override or …`), so a vacant-year scenario can't be modelled	`analyse()`	S
26	LOW	The password gate has no lockout: 1 s per guess, unlimited guesses. Fine on 127.0.0.1, weak if opened to a network.	`dashboard/login.py`	S
27	INFO	104 broad `except Exception` handlers and 3 silent `except: pass` (in the Nehnutelnosti scraper and a dev script)	codebase	—
28	INFO	315 deprecation warnings per test run, all from inside Streamlit (pandas 4). Already handled by pinning pandas below 4.	dependency	—

29	INFO	The pinned Playwright 1.64.1 wants Chromium build 1248. This review ran the browser tests on build 1243 and they passed; CI and Docker install the right build themselves.	`requirements.txt`	—

The three to act on first: (1) Have an účtovník confirm the 2026 Slovak tax figures (findings 2 and 6) and add the 2027 tables in December (finding 3). (2) Decide with a lawyer or notary how to treat the § 15 statutory lien and transfer bans, then fix the LV rules (findings 1 and 4). Until then, keep an Anthropic key set and read every LV yourself before an offer. (3) Compare s.r.o. and personal at the same loan-to-value (finding 5).

2. Everything the dashboard can do — the full capability map

Sovereign is a private tool that finds flats for sale in Slovakia and Czechia priced below the local market. It works out their cash flow under each country's tax and lending rules, screens their title deeds for debts, and tracks the ones worth pursuing through to a purchase-contract draft. It runs as **two processes on one SQLite database** (`data/sovereign.db`): the **dashboard** (Streamlit, `app.py`) and the **scheduler** (`scheduler.py`, every day at 06:00 Bratislava time).

2.1 How you run it

Way	What it starts	Notes
Windows + Docker (`START.bat`, `docker compose up -d --build`)	Dashboard and scheduler containers	Port bound to 127.0.0.1:8501. Database on the named volume `sovereign_data`; logs and backups on your disk.
Mac without Docker (`Start Sovereign.command`)	Dashboard only	Use ▶ RUN EVERYTHING by hand, as there is no scheduler. Accepts Python 3.11–3.14.
Linux / manual (`restart.sh`, `streamlit run app.py`, `python scheduler.py`)	Either	Bound to 127.0.0.1.
Password gate	`DASHBOARD_PASSWORD` in `.env`	Needed before opening it to a network

2.2 Markets

	🇸🇰 Slovakia	🇨🇿 Czechia
Sale portals scraped	Nehnutelnosti.sk (headless Chromium, past its firewall), Bazoš.sk, Topreality.sk	Bezrealitky.cz, Reality.iDNES.cz, Bazoš.cz (Sreality.cz is left out on purpose: its robots.txt disallows crawlers)
Rental portals (live rent comps)	Bazoš, Topreality	Bazoš, Bezrealitky
Currency	EUR	CZK. Converted to EUR at the ČNB rate, fetched daily, with a warning when it's over a week old.
Price benchmark	Regional median €/m², room-adjusted, per okres in Bratislava	ČSÚ purchase prices (kraj / okres), live Prague and Brno district levels
Rent baseline	`config.RENT_PER_M2` (≈110 towns and districts)	Deloitte Rent Index: Prague, Praha 1–10, the 13 regional capitals
Title deed (LV)	kataster.skgeodesy.sk (unofficial scraper)	ČÚZK API KN (free key) plus a pasted výpis z LV
Property tax	Per-town rate per m², and per cadastral area where a town sets one; updated from towns' own sites and CUET	Size coefficient by population × local coefficient from each ordinance (6,258 municipalities)
Lending rules	NBS: 80 % LTV, 70 % for a 3rd+ property from 1 Oct 2026; +2 pp stress test	ČNB: 80 %, 70 % for an investment mortgage from 1 Apr 2026
Contract draft	Kúpna zmluva for a notár	Kupní smlouva for an advokát

2.3 The daily pipeline (14 steps in Slovakia, 13 in Czechia)

#	Step	What it does
1	Backup	An online-API copy of the database (`backups/sovereign-YYYY-MM-DD.db`) and a readable JSON of everything you typed. The newest 14 of each are kept.
2–4	Portal scrapes	Up to 10 pages per portal. Records each portal's count (`portal_runs`). A portal that returns 0 fails its step, and its listings are kept rather than retired.
5	Fill missing details	Re-reads up to 100 Nehnutelnosti listings missing a price, size or district, and retires the ones taken down
6	Rent comps	Scrapes rental listings into live €/m² (or Kč/m²) per district. Applies a 5 % asking-to-achieved haircut, blends with the baseline by sample size (5+ needed), and re-queues affected scores.
7	Housekeeping	Retires listings unseen for 21 days, per portal and only for portals read this run. Flags developer projects.
8	LV debt filter	Screens the title deed's part C for liens, executions, insolvency, lawsuits, auctions, easements and pre-emption rights. Each listing gets PASS, REJECTED or UNVERIFIED; it is never clean without the flat's own LV. Claude (optional) reads the LV too.
9	Address normalisation	Cleans district and city (optionally with Claude)
10	Description enrichment	Reads the ad text for parking, balcony, furnishing, elevator, condition and renovation (rules, plus Claude when a key is set)
11	Cash-flow scoring	Runs the financial engine on every unscored listing
12	Location IQ	Geocodes (Google or Nominatim) and scores transit, amenities, flood (Q100), noise and construction from open data
13	Risk backfill	Fills in risk flags for listings scored before the risk sources existed
14	Merge portal copies	Links the same flat across portals and shows one copy: the LV-verified one, otherwise the cheapest

Scheduling and safety: catch-up runs after downtime, 3 hourly retries when every portal fails, one shared lock so the dashboard and scheduler never run the same step twice (with heartbeat and stale-lock takeover), and status saved to `data/pipeline_status*.json` and shown in red when a run failed. Each market runs on its own, so one failing doesn't stop the other.

2.4 The financial engine (`engine/financial.py`)

- **Rent estimate:** live comps or baseline × room multiplier × parking, furnished, balcony and industrial-zone premiums. Says when it falls back to a regional or national figure.
- **Costs:** mortgage (annuity, split into interest and principal), HOA or SVJ fees, per-town property tax, 5 % vacancy, 5 % owner reserve, optional management fee, 1 % acquisition costs (CZ: plus fixed fees).
- **Two ownership structures, side by side:** personal (SK § 6(3): €500 exemption, no health levy; CZ § 9: flat-rate or actual expenses) and s.r.o. (company loan at +1 pp and 70 % LTV, €1,200/yr running cost, 10 %/21 % corporate tax plus 10 % dividend tax, €2,500 setup with a break-even check).
- **Outputs:** NOI, cap rate, gross and net yield, cash-on-cash, total ROI, self-funding ratio, monthly surplus under each structure, recommended structure, sanity-floor warning.
- **Class:** GREEN at 20 % or more below the regional median €/m², YELLOW at 10 % or more, WHITE otherwise or with no benchmark. Plus a **composite deal grade A–D** from yield, discount, location, energy class, condition and risk.
- **Max offer** 🟡 / 🟢: the highest price that keeps the class. **Rate shock** at +2 pp. **10-year IRR** with appreciation, rent growth, exit costs and exit tax, plus the equity multiple.

- **Tax per year** (`TAX_YEAR_RULES`) with a dated source note. Scores are redone automatically when the tax year, a price, a district's rent or a town's tax changes.

2.5 The dashboard: page header

- **Market switch:** All markets (ranked in EUR), 🇸🇰 Slovakia (EUR), 🇨🇿 Czechia (Kč).
- **KPI tiles:** Total, Green, Yellow, White, Rejected, Pending.
- **Banners:** price cuts in the last 14 days (and whether the new price is inside the max-YELLOW offer), a portal gone quiet (0 listings or failed on its last 2 runs), Czech listings waiting for a rent figure, stale tax year, old FX rate, failed run.

2.6 The 11 tabs, in 3 groups

Group	Tab	What you can do there
🎯 DEALS	TRIAGE TABLE	Every scored listing, sortable. Grade, score, title, district, stage, market, price, Max 🟡/🟢, To 🟡, below-market %, rent, surplus, +2 pp stress, gross yield, cap rate, IRR, days tracked, price change, portals, vibe, LV status, ⛔ cash-flow-negative flag.
	ACTIVE SNAG LIST	GREEN and YELLOW deal cards, 25 a page per class. Full cost breakdown, personal vs s.r.o., financing stress, location risks with sources, price history, other portals' copies, deal stage, vibe notes, **PDF memo download**, LV status and a flat-LV re-check.
	MAP	Every listing with coordinates, coloured by class (faded where only the area is known). The tooltip shows price, class and surplus.
	SATELLITE VIEWER	The listing photo against Google satellite and Street View, and a vibe score with saved notes
📊 ANALYSIS	WHAT-IF / TAX	Re-run any listing or a custom one with your own price, size, district, rooms, LTV, rate, term, s.r.o. premium, LTV and running cost, rent, hold, growth and exit costs. Personal vs s.r.o. table, rate-shock table, equity cash flows chart, IRR, max offers.
	RENT COMPS	Live rent per m² against the baseline per district or town, sample sizes, and the Czech towns waiting for a rent
	TAX RATES	Each Slovak town's rate (engine vs the town's own site), doubtful readings to accept or reject, ordinance PDF upload for blocked towns. Czech coefficients with their sources, and a lookup by place. One click re-reads both countries.
🗂 WORKFLOW	DEAL PIPELINE	A board from WATCHING to VIEWING, OFFER, NEGOTIATING, DUE DILIGENCE, NOTARY, then CLOSED or PASSED, with notes and a timeline per deal
	LV TO-DO	GREEN and YELLOW deals whose title deed isn't verified, and what the plot check found. Bulk entry of the flats' LV numbers, with save, or save and verify now.
	REJECTED	Every LV rejection with its reason, detail and Claude's read. Filter by reason, and re-verify.
	ONE-CLICK CLOSE	A pre-filled SK kúpna zmluva or CZ kupní smlouva draft (buyer, ownership form, price, deposit, escrow, notary) to download. The deal's internal numbers are a separate download.

2.7 Sidebar

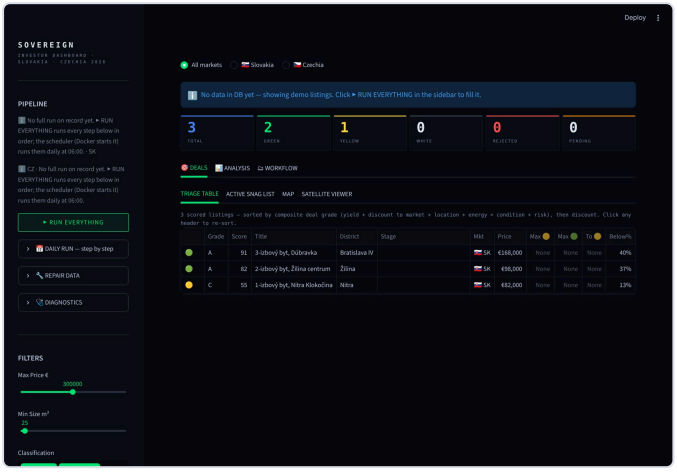
- ▶ **RUN EVERYTHING**: the whole pipeline for each market, in the background with a progress bar.
- 📅 **DAILY RUN — step by step**: one button per portal (6) and per step (Fill missing details, Rent comps, LV debt filter, Normalize addresses, Parse descriptions, Cashflow score, Location IQ, Risk backfill, Merge portal copies).
- 🔧 **REPAIR DATA**: ♻️ Rescore all, 📝 Re-parse (new fields), 🏛️ Update tax rates, 🧹 Clean stale (21 days).
- 🩺 **DIAGNOSTICS**: 🔗 Test sites, 🔗 Test CZ sites (checks reachability and robots.txt).

- **Filters**: max price (€), min size, classification, portals, condition, district search, price drops only, hide developer projects, hide duplicates, require elevator, furnished or parking, cash-flow-positive only, demo data, and an s.r.o. toggle (which structure's figures show).
- **Status line**: last full run, what it found, failed steps, the running job.

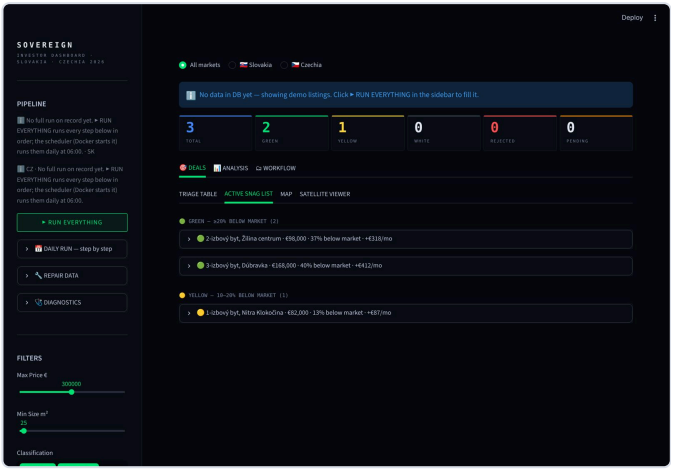
2.8 Documents, data and maintenance

- **PDF investment memo** (`modules/memo.py`, fpdf2), one page per deal. **Contract drafts** (SK and CZ). **Backups and restore** (`python -m modules.backup [restore file]`).
- **Optional APIs:** Anthropic Claude (description parsing on Haiku, LV analysis on Opus), Google Places (geocoding and satellite), ČÚZK API KN (Czech LV), ScraperAPI (IP-block bypass).
- **Maintenance scripts:** `enrich_pending.py`, `repair_prices.py`, `repair_districts.py`, `backfill_enrichment.py`, `diagnose.py`, `seed_market_data.py`, and `dev/` builders for the Czech tax and price data.
- **Quality gates:** 3,734 tests (92 files), ruff (E9, F, B, PLW), CI on Linux (Python 3.11, 3.12, 3.14) and Windows (3.12), a browser job running the scraper in Chromium against a fake site, Dependabot for dependencies and Actions, and a check that `requirements.txt` resolves.
- **Model-input overrides in `.env`** (45 settings in `.env.example`): `PROPERTIES_OWNED`, `SRO_*`, `CASHFLOW_MIN_SURPLUS`, `TAX_YEAR`, `CZ_*`, `BACKUP_KEEP`, `LV_SOFT_FLAGS_REJECT`, the FX rate pin and others.

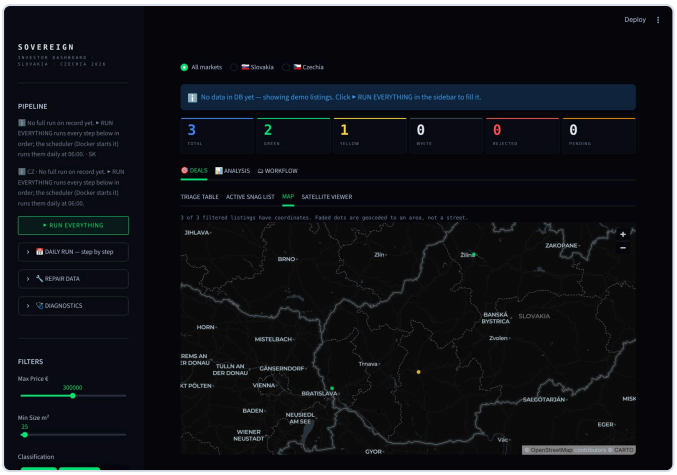
2.9 Screenshots (demo data, run during this review)



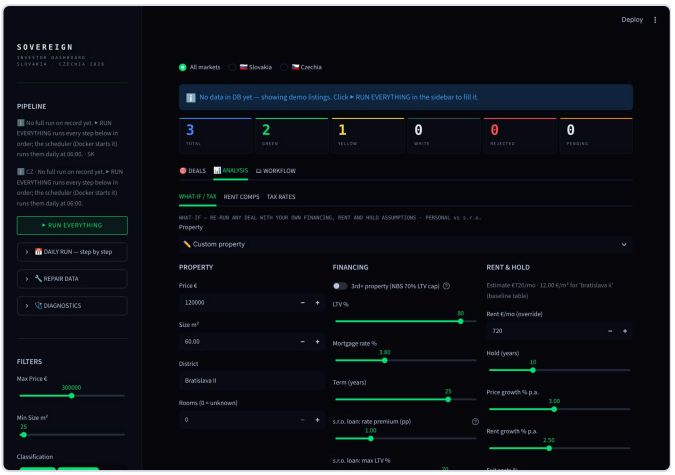
DEALS → TRIAGE TABLE, with the grouped sidebar. Note the grey “None” cells (finding 17) and the “Deploy” button (finding 18).



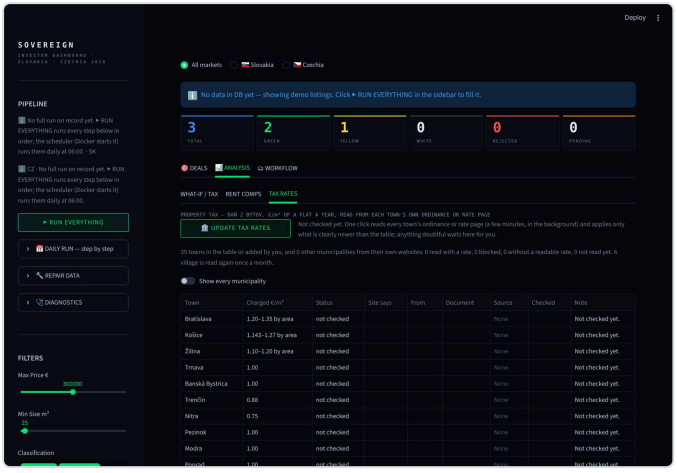
DEALS → ACTIVE SNAG LIST



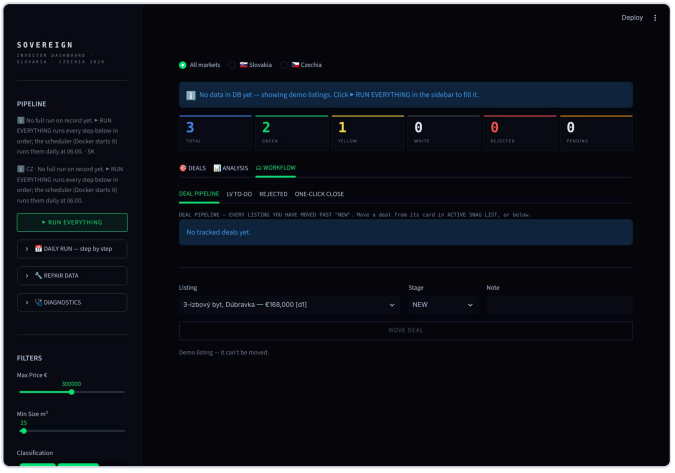
DEALS → MAP



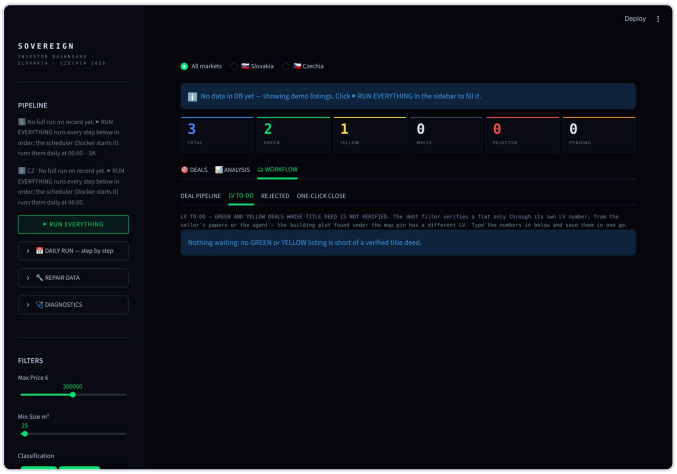
ANALYSIS → WHAT-IF / TAX



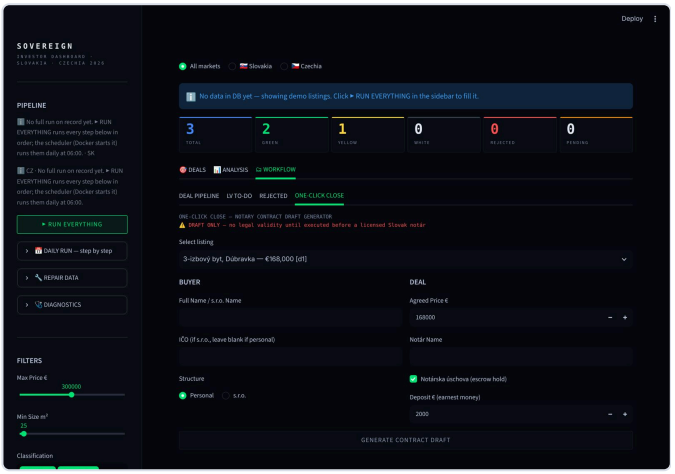
ANALYSIS → TAX RATES



WORKFLOW → DEAL PIPELINE



WORKFLOW → LV TO-DO



WORKFLOW → ONE-CLICK CLOSE

3. Test and check results

Check	Command / method	Result
Full test suite (CI's install, without Playwright)	`pytest tests/` · Python 3.13.18	**PASS** 3,726 passed, 1 skipped (the browser module, run below), 0 failed · 79.7 s
Browser scraper tests (CI's "browser" job)	`SOVEREIGN_BROWSER_TESTS=1 pytest tests/test_nehnutelnosti_browser.py` · Playwright 1.64.1	**PASS** 8 passed · 12 s. The pre-installed Chromium build 1243 stood in for the pinned 1248.
Lint (CI's rule set)	`ruff check .` (E9, F, B, PLW)	**PASS** All checks passed
Dependencies resolve	`pip install --dry-run -r requirements.txt`	**PASS**
Every tab in a real browser	Streamlit on 127.0.0.1, Chromium clicks all 3 groups and 11 tabs	**PASS** No Python exception, no JavaScript error
Phone width (390 px)	Page width check	**PASS** No sideways scroll
Hard-coded secrets	`git grep` for API-key patterns	**PASS** None found
SQL built with f-strings (33 places)	Read each one	**PASS** Only placeholder lists and fixed fragments; user text always goes in as a parameter
Raw HTML with scraped text	Read the 80 `unsafe_allow_html` calls	**PASS** Scraped and typed text goes through `esc()`
Engine edge cases	Script: zero or negative price, zero size, blank district, 0 % rate, 0-year term, 100 % LTV, CZ towns	**2 crashes** at 0 % rate and 0-year term, both unreachable from the UI (finding 15); a LookupError for unknown Czech towns, which is by design (finding 9)
LV screen adversarial texts	Script: 15 crafted part-C entries	**3 wrong** A transfer ban passes (finding 1), the statutory lien rejects (finding 4), "BANKA PLUS s.r.o." clears (finding 22)
Broader static scan	ruff S, BLE, DTZ, SIM; vulture	**INFO** 104 broad excepts, 3 silent passes, no dead code above 80 % confidence
Warnings	pytest warnings summary	**INFO** 315 × Pandas4Warning from inside Streamlit

4. Findings in detail, ranked by importance and severity

Each has a suggested fix. Nothing was fixed in this pass, as you asked. Line numbers are as of commit `79e8edd`.

High

1 · The LV screen misses transfer bans on Slovak title deeds HIGH

`modules/lv_screen.py` · `_RULES` (lines 67–96) · Effort S · ✔ reproduced

Problem The Slovak rules know liens, executions, insolvency, lawsuits, auctions, court bans on disposal (“zákaz nakladania”), seizures, easements and pre-emption rights. They know nothing about **“zákaz scudzenia”, “zákaz prevodu (a zaťaženia)” or “obmedzenie prevodu”**. The Czech screen does: it has “zákaz zcizení”.

Evidence Part-C entries “Zákaz prevodu a zaťaženia v prospech Mesto Žilina” and “Obmedzenie prevodu nehnuteľnosti v prospech ŠFRB” both returned **no hits**. With no other entry, the flat is screened clean.

Impact This is the dangerous direction. A flat that can't legally be sold to you without someone's consent (a town, the state housing fund, a former owner) can show as LV PASS. Claude, when a key is set, may catch it; the keyword screen never will.

Fix Add a rule, e.g. `("zákaz scudzenia", "encumbrance", r"zakaz\w*\s+(?:scudz|prevod|zatazen)\w*|obmedzen\w*\s+prevod\w*")`, as a hard stop. Add tests for each wording.

2 · The 2026 Slovak personal income tax table is out of date HIGH

`config.py` · `TAX_YEAR_RULES[2026]` (lines 81–117) and `calc_income_tax_personal()` · Effort S–M

Problem The table models two bands, 19 % up to €41,445 and 25 % above. Its own comment says the threshold is “carried over from 2023”. The 2026 consolidation package brought in **four bands**: 19 % up to about **€43,983** (154.8 × the subsistence minimum of €284.13), 25 % up to about €60,349, 30 % up to about €75,010 and 35 % above (PwC, Forvis Mazars, Orbitax). The engine also taxes the rental **as if it were your only income**. In reality it stacks on top of your salary, so its marginal rate is your top band.

Impact For anyone whose total income passes about €44k, personal tax on the rental is understated (modelled at 19 %, really 25–35 %). That makes the Personal structure look better than it is, and the surplus too high.

Fix Replace the two bands with a list of bands per year. Add an `OTHER_ANNUAL_INCOME` setting to `.env` and tax the rental at the margin above it. Have an účtovník confirm the figures (see manual task A).

3 · The tax tables expire on 1 January 2027 HIGH

`config.py`, `markets.py` (Czech `tax_year_rules`), `engine/property_tax_sk.py` and `property_tax_cz.py` (`TABLE_YEAR = 2026`) · Effort M every year · open since the previous review's H4

Problem Only a 2026 table exists for each country. From January, scoring falls back to the 2026 rules and the dashboard warns. The figures will then be wrong wherever 2027 rules changed: brackets, the subsistence minimum, the Czech inflation coefficient (§ 11f), NBS and ČNB limits, mortgage rates, and town ordinances effective 1 January.

Fix Add 2027 entries in December, run 🏛️ UPDATE TAX RATES in early January, then ♻️ RESCORE ALL. Steps are in manual task B.

4 · The common statutory owners' lien (§ 15 of Act 182/1993) rejects good flats HIGH

`modules/lv_screen.py` (lien handling, `_BANK_RE`) · Effort S · ✔ reproduced

Problem Many Slovak flat LVs carry “Záložné právo v zmysle § 15 zák. č. 182/1993 Z. z. v prospech vlastníkov bytov a nebytových priestorov v dome”. This lien arises by law and secures the other owners' claims for unpaid building charges. It is not a private debt. The screen reads the creditor (“vlastníkov bytov…”), finds no bank and **blocks**. The repository has no rule or test for it.

Impact Without an Anthropic key, every such flat lands in REJECTED and never reaches your snag list. Silent lost deals, possibly a large share of older blocks. With a key, Claude may clear it, but only when the whole of part C was read.

Fix Recognise “§ 15 … 182/1993” and “v prospech (ostatných) vlastníkov bytov” as a soft flag that is shown but not rejected. Confirm the rule with a lawyer or notary first (manual task C).

Medium

5 · The s.r.o. and personal structures are compared at different leverage MEDIUM

`engine/financial.py` · `analyse()` lines 722–758 · Effort S · ✔ reproduced

Problem The s.r.o. borrows at 70 % LTV and personal at 80 %. A smaller loan means a smaller mortgage payment, so the s.r.o.'s monthly surplus is boosted by the **extra €11–15k of your own cash** it puts in, and that cash is counted as free. `annual_sro_saving` then decides the recommendation.

Evidence Petržalka, €150k, 55 m²: s.r.o. “saving” **+€104/yr** as modelled; at equal 80 % LTV **−€899/yr**. Košice: −€208 → −€865. Žilina: −€130 → −€861. Every time, about €700–1,000/yr of the “saving” is leverage, not tax.

Impact In borderline cases (a saving over about €250/yr pays back the €2,500 setup within 10 years) the tool can recommend founding a company because you put more cash in, not because the tax is lower.

Fix Compare at equal equity: charge the extra equity an opportunity cost (e.g. the mortgage rate). Or compare returns on cash (cash-on-cash or IRR per structure) rather than raw surplus.

6 · Personal tax doesn't deduct mortgage interest MEDIUM

`engine/financial.py` · `personal_rental_base()` lines 256–278 (Slovak branch) · Effort S, plus an adviser's answer

Problem For Slovakia the taxable base is NOI minus €500, and interest is not deducted. Slovak guidance says that for § 6(3) rental, provable expenses including **interest on the loan used to buy the rented flat** are deductible (it cites § 19(3)(t)). Some sources also restrict property tax and insurance when the flat isn't business property.

Impact Personal tax overstated by roughly 19–25 % of the yearly interest (≈€3,500/yr on a €120k loan, so ≈€650–900/yr). This pushes the other way from finding 2. Both need fixing together.

Fix Once an účtovník confirms (manual task A), deduct interest, apply the € 500 exemption with the proportional expense reduction, and list exactly which expenses count.

7 · Headline return metrics always use the s.r.o. scenario MEDIUM

`analyse()` lines 736–744 (“on the s.r.o.'s own financing, as before”) · Effort S

Problem `cash_on_cash`, `net_rental_yield`, `total_roi` and the principal paydown are computed from the s.r.o. numbers. In every probe, Personal was the recommended structure.

Fix Store both sets, and show the one for the recommended structure (or follow the sidebar's s.r.o. toggle). Label which structure a figure belongs to.

8 · What-If: the hold-period slider doesn't move the “Best structure” verdict MEDIUM

`dashboard/tabs/what_if.py` → `analyse()` (`sro_pays` uses `config.HOLD_YEARS`) · Effort S

Problem Set Hold to 3 years: the IRR table updates, but “Best structure” still asks whether the s.r.o. repays its setup within 10 years. The rate-shock table is also titled “(s.r.o.)” even when Personal is recommended.

Fix Pass `hold_years` into `analyse()` and use it in `sro_pays` and the recommendation text.

9 · Czech coverage: sizeable towns wait unscored MEDIUM

`engine/rent_cz.py` · Effort S–M · ✔ reproduced (Kladno → “no rent figure … yet”)

Problem H2 from the previous review is mostly fixed (Praha 1–10 and the 13 regional capitals, rentals read before scoring, a waiting banner). Towns outside the index still need 5 scraped rentals: Kladno, Mladá Boleslav, Opava, Frýdek-Místek, Karviná, Havířov, Teplice, Děčín, Chomutov, Most, Příbram, Kolín and others. A lot of Czech stock sits in exactly these towns.

Fix Allow a sourced kraj-level fallback (the Slovak market already has `_kraj_rent_fallbacks`) with the ⚠ fallback warning. Or add figures from the government's “Cenová mapa nájemného”. Or lower the comp threshold for towns near a regional capital.

10 · Without an Anthropic key, LV decisions rest on keyword rules alone MEDIUM

`modules/debt_bot.py` `_decide_lv()` · Effort: configuration

Problem Claude's read is the second opinion that can catch what the keyword screen misses (finding 1) or clear what it blocks wrongly (finding 4). The key is optional, and without it the screen is the only judge.

Fix Set `ANTHROPIC_API_KEY` (manual task D). Show a dashboard notice when LV checks run without it.

11 · No off-site backup MEDIUM

`modules/backup.py` → `backups/` on the same computer · Effort S (manual)

Problem Daily copies are good: they use SQLite's online backup and keep 14 days. But a disk failure, theft or ransomware takes the database and every backup together. Your deal notes, stages and LV numbers exist nowhere else.

Fix Sync `backups/` to a cloud drive or external disk, and test a restore (manual task H).

12 · Personal data in `cadastre_cache` is kept with no time limit MEDIUM

database · Effort S

Problem Owner names, addresses and full LV texts are stored indefinitely, and nothing purges them. Even for private use, GDPR's storage-limitation principle applies once data concerns other people. It matters much more if the subscription idea goes ahead.

Fix Delete cached LV texts for listings inactive for more than N days (e.g. 180), keeping only the decision and the reason.

13 · Seven Slovak towns still use the property-tax fallback MEDIUM

TAX RATES tab · previous review M2 · Effort S (manual)

Problem Martin, Považská Bystrica, Nové Zámky, Levice, Lučenec, Topoľčany and Rimavská Sobota block automated reads and publish nothing on CUET, so their flats pay the €1.35/m² fallback. In Czechia, 231 ordinances are scans or .doc files the updater can't read.

Fix Run the update from a home connection, or download each town's VZN PDF and upload it on the TAX RATES tab (manual task F).

Low and Info

#	Finding	Detail and fix
14	Orphaned browsers after a scraper timeout	`dashboard/actions.py:48` : `subprocess.run(timeout=1800)` kills only the Python child, not the Node driver or Chromium processes it started. Fix: `start_new_session=True` and kill the process group on timeout.
15	Division by zero at 0 % or a 0-year term	`calc_mortgage()` (line 186). The UI's minimums (1 %, 5 years) hide it. Fix: at r = 0 the payment is principal ÷ n; reject n = 0.
16	IRR “0.0 %” when it can't be worked out	`what_if.py` : `(irr_p.irr or 0) * 100` . Fix: show “—”.
17	Grey “None” cells in the triage table	Max 🟡, Max 🟢 and To 🟡 for listings with no benchmark (seen in the screenshot). Fix: format as “—”, or hide the column when it's empty.
18	The “Deploy” button is visible	Add `[client] toolbarMode = "minimal"` to `.streamlit/config.toml` .
19	Log file never rotates	`scheduler.py:50–60` uses `FileHandler` . Use `RotatingFileHandler(maxBytes=5 MB, backupCount=5)` .
20	The Docker image runs as root and keeps gcc	Add a non-root `USER` , and drop `gcc` if no wheel needs building (all pinned packages ship wheels for 3.12).
21	Contract draft wording	`contract_draft.py:169–232` : “dle § 56a” (“dle” is Czech; Slovak is “podľa”, and the section number should be confirmed). “vyžaduje notariálne vyhotovenie” and “platnosť podpisom … pred notárom”: a Slovak flat purchase needs officially verified signatures, not a notarial deed. The portal's ad title becomes “Nehnuteľnosť”. Have a notary review the template.
22	“banka” in any name clears a lien	“BANKA PLUS s.r.o.” was cleared. The law protects the word, so this is rare. Hardening: require the a.s. legal form, or exclude s.r.o.
23	A connection and schema checks per saved listing	`upsert_listing` opens a connection (with a WAL pragma) and runs 3 `_ensure_*` checks for every listing. Fix: run the checks once per process and batch per page.

24	Unbatched `IN (…)` list	`database/prices.py:90`. Fine below SQLite's 32,766 limit. Batch at 500 like `title_deeds.py` for consistency.
25	A rent override of 0 is ignored	`rent_override or get_rent_estimate(…)`. Use `is None` if a 0-rent stress scenario is wanted.
26	No lockout on the login	Fine for 127.0.0.1. Before exposing it: a lockout after 5 tries, or put it behind a VPN or tunnel with its own login.
27	Broad exception handlers	104 broad excepts, mostly deliberate “fail soft” guards. Silent `pass` at `scraper/nehnutelnosti.py:741` and `1225`; add a debug log line.
28	pandas 4 deprecation inside Streamlit	Already pinned to pandas < 4 and ignored by Dependabot. Revisit when Streamlit fixes it.
29	Chromium build pin	CI and Docker install build 1248 themselves. Just remember `playwright install chromium` after any Playwright upgrade outside Docker.

5. What happened to the previous review (5 Nov 2026)

Most of it was done within a day. Summary: **15 of 18 issues fixed**, 2 partly done, 1 open. **7 of 14 dashboard ideas done**, 2 partly done, 5 not started.

Item	Status	Notes
H1 No Czech green or yellow	DONE	ČSÚ purchase prices, plus live Prague and Brno district levels (PR #72)
H2 Czech listings never scored	PARTLY	13 capitals and Praha 1–10, rentals read before scoring, a waiting banner. Other towns still wait (finding 9).
H3 A broken portal's listings retired	DONE	Per-portal retirement, a quiet-portal warning, `portal_runs`
H4 Tax tables expire 1 Jan 2027	OPEN	Finding 3
M1 Unverified model inputs	DONE	`docs/M1_VERIFICATION.md`. 12 items remain listed as UNVERIFIED in the README.
M2 Property-tax fallbacks	PARTLY	Zvolen, Liptovský Mikuláš and Čadca added, CUET fallback. Seven towns left (finding 13).
M3 CI never runs the browser scraper	DONE	A browser job against a fake site
M4 Database set-up on every click	DONE	`st.cache_resource`
M5 Listings reloaded on every click	DONE	60 s cache, cleared after jobs
M6 Silent errors	DONE	Logged with the exception type
M7 Very large files	DONE	`app.py` is now 111 lines plus a `dashboard/` package; `database/` is a package
L1 Minimal lint · L2 Python versions · L3 pandas · L4 test-only code · L5 old branches · L6 fixed FX rate · L7 updater housekeeping	DONE	B and PLW rules; 3.11, 3.12, 3.14 and Windows in CI; pandas < 4; helpers removed; only `main` is left; ČNB rate fetched daily; sets and closed sessions
L8 Sreality not scraped	BY DESIGN	robots.txt

Dashboard ideas from the previous review

#	Idea	Status
1	Portal health strip (count today, 7-day trend, last success)	PARTLY a quiet-portal warning exists; no strip with counts and trend
2	Separate cheap from profitable	DONE the ⛔ flag and the cash-flow-positive filter
3	Say why listings are waiting	DONE
4	Group the sidebar · full portal names	DONE
5	Fewer, grouped tabs	DONE 3 groups
6	Column chooser for the triage table	NOT DONE
7	Alerts (email or Telegram)	NOT DONE
8	Side-by-side comparison	NOT DONE
9	A richer map (grade and cash-flow colours, clustering, travel time)	PARTLY coloured by class with a surplus tooltip; no clustering, no cash-flow colours, no travel time

10	Freshness everywhere	PARTLY FX rate daily and stale notices; no “last seen” age on cards
11	CSV or XLSX export	NOT DONE
12	A faster feel	DONE
13	Hide the Deploy button	NOT DONE (finding 18)
14	Compact card view for phones	NOT DONE the page fits at 390 px, but cards are desktop cards

6. Suggestions

6.1 Still open from the previous review (re-ranked)

1. **Alerts**: Telegram or email when a new GREEN, cash-flow-positive listing appears, or a tracked deal's price drops. This is the core promise (“wake up to a shortlist”), and today you still have to open the dashboard. A Telegram bot takes about an hour (manual task I).
2. **Export** the filtered triage table to CSV or XLSX (`st.download_button` with `df.to_csv`), for your bank, adviser or partner.
3. **Side-by-side comparison** of 2–4 listings: price per m², yield, both structures, risks and LV status.
4. **Portal health strip**: count today, a 7-day sparkline and the last success per portal (the data is already in `portal_runs`).
5. **Column chooser** for the triage table (pin Grade, Title, Price and Surplus; `st.dataframe(column_order=…)` with a multiselect).
6. **A richer map**: colour by cash flow, cluster pins, filter by distance or travel time to a point (work, a university).
7. **“Last seen N days ago”** on every card.
8. **Hide Deploy** (one line), and a **compact phone card** (title, price, class, surplus, one button).

6.2 New suggestions

#	Suggestion	Why	Effort
N1	**An “Your tax situation” panel**: other yearly income, owned properties, whether you'd use an existing s.r.o.	Fixes finding 2's stacking problem, and makes Personal vs s.r.o. personal to you	M
N2	**Offer-letter / negotiation sheet**: the max-🟡 price, the comps that justify it, the price history and days on market, as a one-page PDF to take to a viewing	Turns “Max offer” into an argument the seller sees	M
N3	**A due-diligence checklist per deal** in the pipeline: LV checked ≤ 48 h before signing, energy certificate, HOA (správca) confirmation of no arrears, fond opráv balance, building revision reports, utility meter readings	Each item is a standard Slovak or Czech purchase risk the tool can track	S–M
N4	**Ask the správca / SVJ** template: a pre-written email or letter asking for the potvrdenie o bezdlžnosti, fond opráv balance and planned works	The HOA's debts and planned levies are invisible on the LV	S
N5	**Mortgage-offer comparer**: type 2–3 bank offers (rate, fixation, fees, insurance) and see the effect on surplus and IRR	The model uses one average rate; real offers differ by 0.5 pp or more	S–M
N6	**Fixation-end shock**: model the rate resetting after 3/5/10 years (not just +2 pp from day one)	That's how Slovak and Czech mortgages actually reprice	S
N7	**Short-term let scenario** (Airbnb-style) for Tatras, Bratislava I and Praha 1–2, with occupancy and platform fees	Poprad already gets a “Tatry Airbnb premium” in the baseline; make it explicit and switchable	M
N8	**Renovation-flip scenario**: buy price + renovation cost/m² against the post-renovation median, with time and financing costs	Condition is already parsed; many GREEN listings are GREEN because they need work	M
N9	**“Why this grade?” breakdown** on each card: points from yield, discount, location, energy, condition and risk	The A–D grade is a black box today	S
N10	**Watchlist by saved search** (district + size + max price) with its own alert	Lets you monitor a target area, not just GREENs	M

N11	**Portfolio view** for flats you bought: actual rent, costs, tax paid, equity, LTV over time	The tool stops at the contract. Tracking real numbers also calibrates the model's assumptions.	L
N12	**Backtest the classifier**: of last month's GREENs, how many sold, at what discount, and how fast	Tells you whether GREEN really means cheap. The data is already in price history and deactivations.	M
N13	**Weekly digest PDF or email**: new GREENs, price cuts, pipeline changes, portal health	A summary without opening the app	S–M
N14	**Dashboard notice when no Anthropic key is set** and LV checks are keyword-only	Findings 1, 4 and 10	S
N15	**A year-end checklist page** (December: tax tables, NBS/ČNB, ordinances), shown from 1 December	Finding 3 recurs every year	S
N16	**LV rules review mode**: paste any LV and see which entries the screen flags, and why	Lets you and your lawyer check the rules against real LVs	S
N17	**Energy-certificate and heating-cost estimate** by energy class and size	Utilities matter to tenants and to the rent a flat achieves	S–M

7. What you need to do manually — step by step

These need you: a decision, a professional's confirmation, an account or a key, or access from your own computer. Code changes (the fixes in section 4) can be done for you in a follow-up session once you say so.

Task	Why	When	Time
A. Confirm the 2026 Slovak tax figures with an účtovník	Findings 2 and 6	This month	1 h + adviser
B. Collect the 2027 tax and lending figures	Finding 3	1–20 December, then early January	2–3 h
C. Decide the LV rules with a lawyer or notary	Findings 1, 4, 21	Before your next offer	1 h + adviser
D. Set up API keys and `.env`	Finding 10, better data	This week	30–45 min
E. Get real s.r.o. loan and running-cost quotes	Finding 5, M1 items	Before choosing a structure	1–2 h
F. Upload the blocked towns' tax ordinances	Finding 13	This month, again in January	1 h
G. Feed LV numbers for your GREEN and YELLOW deals	Turns UNVERIFIED into PASS or REJECTED	Per deal	5 min per deal
H. Set up an off-site backup and test a restore	Finding 11	This week	30 min
I. (Optional) Create a Telegram bot for alerts	Suggestion 6.1-1	Whenever	15 min
J. Keep the app updated and healthy	Dependabot, Docker	Weekly	10 min
K. Legal advice before selling access	Subscription or assignment idea	Before launch	Adviser

A · Confirm the 2026 Slovak tax figures with an účtovník

Resources: podpora.financnasprava.sk (Finančná správa help: search “príjmy z prenájmu § 6 ods. 3”) · slov-lex.sk (Act 595/2003 Z. z., current version) · PwC Worldwide Tax Summaries: Slovakia · Forvis Mazars CEE tax guide 2026

1. Find an účtovník or daňový poradca (Slovenská komora daňových poradcov lists members at skdp.sk).
2. Send them these questions (copy them as they are):

```
1. 2026 personal income tax bands and thresholds (19/25/30/35 %?), and the
   subsistence minimum used (€284.13 → €43,983.32?).
2. Rental income under § 6 ods. 3 by a natural person without živnosť:
   a) is interest on the mortgage used to buy the rented flat a deductible
      expense? (§ 19 ods. 3 písm. t?)
   b) are property tax, insurance, fond opráv, správa, repairs and
      depreciation deductible if the flat is not business property?
   c) how exactly is the €500 exemption (§ 9 ods. 1 písm. g) applied, and
      how are expenses reduced proportionally?
   d) are health contributions due on this income in 2026? (model: no)
3. s.r.o.: 2026 corporate rates (10 % up to €100k revenue / 21 % / 24 %),
   dividend withholding tax (10 %?), any minimum tax (daňová licencia).
4. Exit: is a personal sale tax-free after 5 years of ownership in 2026?
```

3. Write their answers next to `TAX_YEAR_RULES[2026]` in `sovereign_final/config.py`, with the date and their name as the `source`. Or send them to Claude to change.
4. Click ♻️ **RESCORE ALL** in the sidebar (🔧 REPAIR DATA), so every listing is scored again.
5. Repeat the same questions for Czechia with a daňový poradce if you buy there (2026 Czech brackets, § 9 flat-rate expenses, SVJ deductibility).

B · Collect the 2027 tax and lending figures (every December)

Resources: financnasprava.sk · slov-lex.sk · employment.gov.sk (životné minimum, usually set from 1 July) · nbs.sk (macroprudential policy: housing-loan limits) · cnb.cz (LTV/DTI limits) · financnisprava.cz · zakonyprolidi.cz (586/1992 Sb., 338/1992 Sb.) · czso.cz (CPI for the § 11f inflation coefficient)

1. **1–10 December:** on slov-lex.sk open Act 595/2003 and switch to the version effective 1.1.2027. Note the bands, the subsistence minimum, the corporate and dividend rates, and the € 500 exemption.
2. On nbs.sk, check whether the LTV caps changed (80 %, 70 % for a 3rd+ property) and the +2 pp stress add-on. On cnb.cz, check the Czech limits.
3. Note a current average mortgage rate for both countries (NBS statistics; Swiss Life Hypoindex for Czechia).
4. Czechia: the inflation coefficient for 2027 (CPI May 2026 over May 2023, rounded down) is published by the Financial Administration (financnisprava.cz). Write it down.
5. Ask Claude to add the `2027` entries to `config.TAX_YEAR_RULES`, the Czech rules in `markets.py` and the property-tax `TABLE_YEAR`, giving it your numbers and sources.
6. **2–10 January:** click 🏛️ **UPDATE TAX RATES** (towns' new ordinances take effect on 1 January), review doubtful readings on the TAX RATES tab, then ♻️ **RESCORE ALL**.
7. Check that the dashboard no longer shows the “stale tax year” notice.

C · Decide the LV rules with a lawyer or notary

Resources: notar.sk (Notárska komora SR) · sak.sk (Slovenská advokátska komora) · katasterportal.sk / kataster.skgeodesy.sk (ÚGKK) · Act 182/1993 Z. z. on slov-lex.sk · Czech: nahlizenidokn.cuzk.cz

1. Print or export 3–5 real LVs from flats you looked at (katasterportal.sk; a free informative extract is enough).
2. Ask the lawyer or notary:

```
1. "Záložné právo podľa § 15 zák. 182/1993 v prospech vlastníkov bytov":
   harmless for a buyer if the HOA confirms no arrears (potvrdenie o
   bezdlžnosti)? Should it be a soft flag rather than a reject?
2. "Zákaz scudzenia / zákaz prevodu / obmedzenie prevodu" (e.g. in favour
   of a town or ŠFRB): always a hard stop, or can it be cleared with the
   holder's consent?
3. Which other part-C entries should always stop a purchase?
4. Review the kúpna zmluva template (ONE-CLICK CLOSE): legal references
   (notárska úschova section), wording, what must be added.
```

3. Write the decisions down. Ask Claude to change `modules/lv_screen.py` to match (findings 1 and 4), with tests for each wording.
4. Until then: check every LV yourself on REJECTED (“reason: lien, creditor ‘vlastníkov bytov…’”) and before any offer.
5. Always re-verify the LV within 48 hours of signing.

D · Set up API keys and the `.env` file

Resources: console.anthropic.com (API keys, billing) · console.cloud.google.com (Places / Maps APIs) · registrace.cuzk.gov.cz (free ČÚZK API KN key) · the comments in `sovereign_final/.env.example`

1. In `sovereign_final/`, copy `.env.example` to `.env` if you haven't. It is gitignored, so keys never reach GitHub.
2. **Anthropic (recommended):** create a key at console.anthropic.com → API Keys, set a monthly spend limit under Billing, then add `ANTHROPIC_API_KEY=…`.
3. **Google (optional):** create a project, enable “Places API” and “Maps Static API” or “Maps JavaScript API”, create an API key, restrict it to those APIs, then add `GOOGLE_PLACES_API_KEY=…`.
4. **ČÚZK (Czech deals):** register at registrace.cuzk.gov.cz for the API KN, then add `CZ_CUZK_API_KEY=…`.
5. Set `PROPERTIES_OWNED=` to how many residential properties you own (2 or more switches to the 70 % LTV cap). Set `NOMINATIM_EMAIL=` to your email (OpenStreetMap's usage policy asks for it).
6. Optional: `CASHFLOW_MIN_SURPLUS=50` (only count a deal positive above €50/month), and `DASHBOARD_PASSWORD=` a long password, but only if you'll open it from another device.
7. Apply it: `docker compose up -d` (no rebuild needed), or restart `Start Sovereign.command`.
8. Click 🔒 **LV DEBT FILTER** and 📝 **RE-PARSE** so existing listings get the Claude read.

E · Get real s.r.o. loan and running-cost quotes

Resources: your bank's business desk (Tatra banka, VÚB, SLSP, ČSOB, Prima banka) · your účtovník · `.env` keys `SRO_RATE_PREMIUM_PP`, `SRO_LTV_RATIO`, `SRO_ANNUAL_RUNNING_COST`

1. Ask 2 banks for an indicative loan to a new one-flat s.r.o.: rate, maximum LTV, fees, and whether a personal guarantee is required.
2. Ask your účtovník for a yearly fixed price for a one-flat s.r.o. (bookkeeping, accounts, tax return), plus the registered office and bank account costs.
3. Put the numbers in `.env`, e.g. `SRO_RATE_PREMIUM_PP=0.012`, `SRO_LTV_RATIO=0.7`, `SRO_ANNUAL_RUNNING_COST=1100`. Restart, then ♻️ **RESCORE ALL**.
4. Until finding 5 is fixed, compare the structures in WHAT-IF with the s.r.o. max LTV slider set to the same LTV as personal.

F · Upload the blocked towns' tax ordinances

Resources: each town's website, “Úradná tabuľa” or “VZN” section (search “VZN daň z nehnuteľností 2026 <town>”) · cuet.slovensko.sk (central notice board)

1. First try 🏛️ **UPDATE TAX RATES** from your home connection: town sites often block data-centre IPs only.
2. For each town still on the fallback (Martin, Považská Bystrica, Nové Zámky, Levice, Lučenec, Topoľčany, Rimavská Sobota), open its website and find the **VZN o miestnych daniach** (or **o dani z nehnuteľností**) in force for 2026. Download the PDF.
3. In the dashboard: 📊 ANALYSIS → **TAX RATES** → the town → upload the PDF. Accept the reading if the rate per m² for “byty” matches the PDF.
4. Repeat in January for 2027 ordinances.

G · Feed LV numbers for your GREEN and YELLOW deals

Resources: the seller or agent (the LV number is on the seller's title documents) · katasterportal.sk

1. Open 🗂 WORKFLOW → **LV TO-DO**. It lists the GREEN and YELLOW deals whose own LV isn't read.
2. Ask the agent for “číslo LV bytu a katastrálne územie”. Agents usually have it.
3. Type the numbers into the table (bulk entry), then click **save & verify now**.
4. Check the result on the card (PASS / REJECTED, with the reason). For a REJECTED one, read the LV yourself before you discard the deal (finding 4).

H · Set up an off-site backup and test a restore

Resources: Google Drive, OneDrive or iCloud desktop sync, or an external USB disk · README “Backups”

1. Pick a synced cloud folder, e.g. `OneDrive\Sovereign-backups`.
2. **Windows:** Task Scheduler → Create Basic Task → daily 07:30 → Start a program: `robocopy` with arguments `"C:\path\to\sovereign_final\backups" "C:\Users\you\OneDrive\Sovereign-backups" /MIR`.
   Mac: Automator or a `crontab -e` line: `30 7 * * * rsync -a ~/path/sovereign_final/backups/ ~/Library/Mobile\ Documents/com~apple~CloudDocs/Sovereign-backups/`
3. Test a restore once (stop the scheduler first):

```
docker compose stop scheduler
docker compose exec dashboard python -m modules.backup restore backups/sovereign-YYYY-MM-DD.db
docker compose start scheduler
```

4. Check that your deal stages and notes are still there.

I · (Optional) Create a Telegram bot for alerts

Resources: Telegram's `@BotFather` · core.telegram.org/bots/tutorial

1. In Telegram, message `@BotFather` → `/newbot` → choose a name. Copy the bot token.
2. Send your new bot any message, then open `https://api.telegram.org/bot<TOKEN>/getUpdates` in a browser and copy your `chat.id`.
3. Put both in `.env` (`TELEGRAM_BOT_TOKEN`, `TELEGRAM_CHAT_ID`) and ask Claude to build the alert step (suggestion 6.1-1).

J · Keep the app updated and healthy (weekly, 10 minutes)

Resources: GitHub → Pull requests (Dependabot) · docs.docker.com/compose

1. On GitHub, merge Dependabot PRs whose CI checks are green.
2. Update locally: `git pull`, then `docker compose up -d --build --remove-orphans`. The rebuild also installs the matching Chromium.
3. Glance at the dashboard header: no red failed-run banner, no ⚠ quiet portal. If a portal is quiet for 2+ runs, click 🔗 **TEST SITES** and tell Claude what it says.
4. Once a month, check the size of `logs/scheduler.log` and delete it if it's large (until finding 19 is fixed).

K · Before selling access or assigning deals for a fee

Resources: a Slovak lawyer (sak.sk) · Úrad na ochranu osobných údajov (dataprotection.gov.sk) · README “Before this becomes a product”

1. Ask about: portal terms and database rights for reusing scraped data; the unofficial cadastre scraper; GDPR (owner names in `cadastre_cache`, finding 12); brokerage licensing and AML duties for assignment fees; whether a ranked paid feed counts as investment advice.
2. Don't launch the €500/month feed or the deal-assignment model until you have written answers.

8. How this review was done, and sources

- **Tests:** a fresh virtual environment with CI's install (requirements minus Playwright, plus pytest and ruff), the full suite run, then the browser module with Playwright 1.64.1 and the pre-installed Chromium.
- **Static checks:** ruff with CI's rules and extra security, exception and datetime rules (S, BLE, DTZ, SIM), vulture for dead code, a secrets grep, and a read of each f-string SQL site and each raw-HTML call.
- **Running the app:** a copy of the tracked files in a scratch folder (empty database, demo listings), Streamlit on 127.0.0.1, Chromium clicking all 3 groups and 11 tabs at 1440 px and 390 px, screenshots.
- **Probing:** scripts against `engine/financial.py` (edge cases, Slovak and Czech towns, s.r.o. at equal LTV) and `modules/lv_screen.py` (15 crafted part-C entries).
- **Reading the code:** config, the financial engine, scheduler, dashboard actions, the database layer (listings, the lock), backup, login, Docker, the contract draft and the LV decision logic.
- **Web check** of the 2026 Slovak tax rules: the sources below. They are secondary sources; confirm with the act and an adviser (task A).

Sources

- PwC Worldwide Tax Summaries — Slovak Republic, individual taxes
- Forvis Mazars — Central and Eastern European tax guide 2026
- Orbitax — Slovak Ministry of Finance release on the 2026 brackets
- Finančná správa — Príjmy z prenájmu nehnuteľností podľa § 6 ods. 3
- Prenájom nehnuteľnosti a úroky z hypotéky v daňovom priznaní
- Previous review: `sovereign_final/docs/PROJECT_REVIEW.md` (5 Nov 2026)

Not covered: live scraping of the real portals and the cadastre (no network from the review machine, and it would hit third-party sites), the real database, and a Windows run (CI covers Windows).